

Ratio and Rate Problem Solving

Lesson 3-7

Name: _____

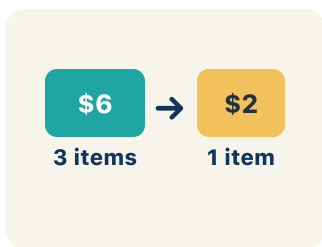
Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.



$\$12 \text{ for } 4 \text{ pounds} \rightarrow \text{dollars per pound}$

Rate

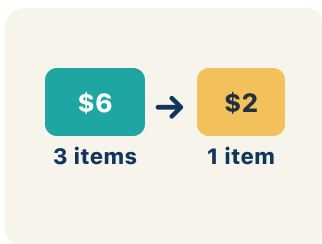
A ratio comparing two amounts with different units, like miles per hour.



$\$12 \div 4 = \$3 \text{ per } 1 \text{ pound}$

Unit rate

A rate for just 1 of something, like cost for 1 item.



$60 \text{ miles per hour}, \5 per ticket

Per

For each one. Example: 5 dollars per book.

247

hundreds tens ones

$\text{Set up} \rightarrow \text{Plan} \rightarrow \text{Solve} \rightarrow \text{Check}$

Problem solving

Using ratios and rates to find a missing amount.

$$\frac{1}{2} = \frac{2}{4}$$

$\frac{3}{5} = \frac{x}{20} \rightarrow x = 12$

Proportion

A math sentence saying two ratios are equal. It helps find a missing number.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can solve real-world rate problems by finding and using unit rates.

1 Notice

It's the final day of Chef Academy — the Culinary Competition!

2 Key idea

I can solve real-world rate problems by finding and using unit rates.

3 Words I'll use

Rate

Unit rate

Per

Problem solving

Proportion



Think About It

- What two quantities with different units are being compared?
- How could you find the cost of just 1 pound of chicken?
- How could you find the number of cupcakes frosted in 1 minute?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A bakery makes 24 cookies in 3 batches. What is the unit rate of cookies per batch?

- A. 8 cookies per batch
- B. 3 cookies per batch
- C. 24 cookies per batch
- D. 6 cookies per batch

Step 1

Unit rate: $24 \text{ cookies} \div 3 \text{ batches} = 8 \text{ cookies per batch}$.



Answer: A. 8 cookies per batch

 We do — together

Solve this one with your class using the same steps.

Problem: Apples cost \$5 for 4 pounds. What is the cost per pound?

- A. \$1.25
- B. \$0.80
- C. \$1.50
- D. \$4.00

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A cyclist rides 36 miles in 3 hours. At the same rate, how far will she ride in 5 hours?

- A. 60 miles
- B. 48 miles
- C. 72 miles
- D. 45 miles

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Team 1 knows 3 pounds of chicken cost \$18. What was your plan to find the cost of just 1 pound, and why is that unit rate useful?

Level 1 support

Start here: Divide the total cost by the number of pounds to find the cost per 1 pound.

 **Need a hint?**

Sentence starters (optional)

My plan was to divide ____ by ____ to get the cost per ____.

Mi plan fue dividir ____ entre ____ para hallar el costo por ____.

The unit rate is ____ per pound, which is useful because ____.

La tasa unitaria es ____ por libra, lo cual es útil porque ____.

Word bank: rate unit rate per proportion divide

Listen for: Students divide \$18 by 3 to get \$6 per pound and explain the unit rate lets them find the cost of any number of pounds.

Level 2

Once you know the unit rate is \$6 per pound, how can you find the cost of 7 pounds without a ratio table? Justify your shortcut.

- I can find the cost of 7 pounds by ____ because ____.
- The unit rate works as a shortcut since ____.

EXPLORE

Team 2 frosts 5 cupcakes every 4 minutes. How did you set up a rate or proportion to find how many cupcakes they frost in a longer time?

Level 1 support

Start here: Use the rate of 5 cupcakes per 4 minutes, or find cupcakes per 1 minute, then scale.

 Need a hint?

Sentence starters (optional)

The rate is ____ cupcakes per ____ minutes.

La tasa es ____ cupcakes por ____ minutos.

To find the total, I ____ the rate.

Para hallar el total, ____ la tasa.

Word bank: rate unit rate per proportion scale

Listen for: Students state the rate (5 cupcakes per 4 minutes) and set up a proportion or unit rate (1.25 cupcakes/min) to scale to the needed time.

Level 2

Would it be faster to use a unit rate or a proportion for this problem? Explain when each strategy is the better tool.

- I would use a ____ here because ____.
- A proportion is better when ____, but a unit rate is better when ____.

CONNECT

Tell about a real situation where rates matter, like speed, pay, or price, and explain how the unit rate helps you.

Level 1 support

Start here: Rates like miles per hour or dollars per pound let us predict and compare amounts.



Need a hint?

Sentence starters (optional)

Rates matter when ____, like ____ per ____.

Las tasas importan cuando ____, como ____ por ____.

The unit rate helps me because ____.

La tasa unitaria me ayuda porque ____.

Word bank: **rate** **unit rate** **per** **proportion** **compare**

Listen for: Students give a real rate (speed, hourly pay, price per item) and explain how the unit rate lets them predict or compare.

Level 2

Pick two real rates that could be compared, and explain how unit rates would let you decide which is the better choice.

- I could compare ____ and ____ by finding ____.
- The better choice would be the one with ____ because ____.

Write About the Math

The Writing Revolution

I can explain my solution using the words rate, unit rate, per, and proportion.

1. Kernel Sentence subject + verb

Model: Rate is a ratio comparing two amounts with different units, like miles per hour.

Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.

Write a kernel sentence about rate. Use a subject and a verb.

Escribe una oración base sobre tasa. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rate matters in math

Tasa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rate matters in math because ____.

Tasa importa en matemáticas porque ____.

but
pero

Rate matters in math, but ____.

Tasa importa en matemáticas, pero ____.

so
entonces

Rate matters in math, so ____.

Tasa importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about rate.
Di un hecho verdadero sobre rate.

Rate ____.

Question *Pregunta*

Ask a question about rate.
Haz una pregunta sobre rate.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about rate.
Muestra entusiasmo sobre rate.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with rate.
Dile a un compañero qué hacer con rate.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

My plan was to divide ____ **by** ____ **to get the cost per** ____.

Mi plan fue dividir ____ *entre* ____ *para hallar el costo por* ____.

The unit rate is ____ **per pound, which is useful because** ____.

La tasa unitaria es ____ *por libra, lo cual es útil porque* ____.

The rate is ____ **cupcakes per** ____ **minutes.**

La tasa es ____ *cupcakes por* ____ *minutos.*

Try It

Solve on your own. Check the answer key when you are done.

1. A runner covers 12 km in 4 hours. How long to cover 21 km at the same speed?

- A. 7 hours
- B. 6 hours
- C. 8 hours
- D. 5 hours

Show your work:

2. 5 notebooks cost \$7.50. What is the cost of 12 notebooks?

- A. \$18.00
- B. \$15.00
- C. \$20.00
- D. \$16.50

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A factory makes 240 widgets in 8 hours. Another factory makes 200 widgets in 5 hours. Which factory is faster? If both factories work together for 10 hours, how many total widgets would they produce?

Show your work:

Reflect — Exit Ticket

A printer prints 30 pages in 5 minutes. At this rate, how many pages will it print in 12 minutes?

A. 72

B. 60

C. 36

D. 42

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. \$1.25 — *Unit rate: $\$5 \div 4 \text{ pounds} = \1.25 per pound .*
2. **You Try (notes):** A. 60 miles — *Unit rate: $36 \div 3 = 12 \text{ miles per hour}$. In 5 hours: $12 \times 5 = 60 \text{ miles}$.*
3. **Try It 1:** A. 7 hours — *Speed is 3 km/h; $21 \div 3 = 7 \text{ hours}$.*
4. **Try It 2:** A. \$18.00 — *Each notebook is \$1.50; $12 \times \$1.50 = \18.00 .*
5. **Exit Ticket:** A. 72 — *Unit rate: $30 \div 5 = 6 \text{ pages per minute}$. In 12 minutes: $6 \times 12 = 72 \text{ pages}$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).